



Ambilpur Vaishnav

Roll No: 24112011182

B.Tech | Computer Science Engineering

Maulana Azad National Institute of Technology, Bhopal

+91-7396806656

vaishnavambilpur2006@gmail.com

GitHub Profile

LinkedIn Profile

EDUCATION

Maulana Azad National Institute of Technology, Bhopal	2024 – 2028
B.Tech in Computer Science Engineering (Expected May 2028)	CGPA: 8.88
TSBIE (Senior Secondary, Class XII)	2024
Telangana State Board of Intermediate Education	97.1%
TSBSE (Secondary, Class X)	2022
Telangana State Board of Secondary Education	98%

PROJECTS

AI-Powered Semantic Reddit Search Engine

[Repo link](#) | [Live link](#)

Personal Project

- Engineered a semantic search engine over Reddit using AI vector embeddings to retrieve results by contextual meaning rather than keywords; combined a third-party Reddit data API with an ArcticShift archive fallback and a Hugging Face cross-encoder reranker, while integrating Redis caching to deliver near-instant repeat-query responses and a self-growing index that autonomously expands with every search.

Real-Time Queue Management SaaS for Businesses

[Repo link](#) | [Live link](#)

Collaborative Project

- Architected a multi-tenant queue management platform enabling businesses to onboard customers instantly via a zero-install browser link, eliminating physical waiting lines through live position tracking and real-time estimated wait-time delivery.
- Optimised real-time event propagation over persistent WebSockets for near-instant queue updates; designed a B2B REST API with a self-provisioning sandbox reducing third-party integration to a single API call; recognised as Top 6 out of 60 teams at Hacksogen, IIIT Gwalior IEEE.

End-to-End Encrypted Real-Time Chat Application

[Repo link](#) | [Live link](#)

Personal Project

- Developed a privacy-first real-time chat application with client-side End-to-End Encryption using AES-GCM and PBKDF2 key derivation via the browser's native Web Crypto API, ensuring complete message confidentiality such that even a fully compromised server exposes zero readable content.
- Optimised frontend state management with Zustand to eliminate redundant re-renders, achieving low-latency live features including typing indicators and user presence across all devices.

SKILLS

- Languages:** C, C++, JavaScript, TypeScript
- Frontend:** HTML5, CSS3, React 18, Next.js, Tailwind CSS, Zustand
- Backend:** Node.js, Express, REST APIs, WebSockets, Socket.io, Prisma ORM
- Databases & Cloud:** MongoDB, PostgreSQL, pgvector, MySQL, Redis, Vercel, Render
- AI / Search:** Vector Embeddings, Semantic Search, pgvector HNSW, Hugging Face
- Tools & Software:** Git, GitHub, Postman, Upstash
- CourseWork:** Data Structures & Algorithms, Database Management Systems, Object Oriented Programming

ACHIEVEMENTS & CODING PROFILES

- LeetCode:** Achieved a Max Rating of **1707** (Top 12% globally); solved **350+** algorithmic problems across data structures and algorithms. [Profile](#)
- GeeksforGeeks:** Solved **150+** DSA problems. [Profile](#)
- Hacksogen (IIIT Gwalior IEEE): Top Performer** (Top 6 out of 60 teams) for Real-Time Queue Management SaaS in the Web Development Track. [Certificate](#)
- Dev Dual Hackathon (IIT Hyderabad):** Ranked **Top 11 out of 120+ teams** in a national-level full-stack competition. [Certificate](#)

POSITIONS OF RESPONSIBILITY

Member, Zenith (The Programming Club, NIT Bhopal)

May 2026 – Present

- Participated in **competitive programming contests** and **hackathons**, solving algorithmic challenges; proficient in **C++** and **DSA**.

Content Writer, Quizzer Club, NIT Bhopal

Jul 2025 – Present

- Managed content for **IQC** and **Anveshan** (**200+ participants**); handled sponsorship reports and social media outreach.